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Department of Earth Sciences
 University of Connecticut
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Education:

Ph.D. 2021, University of Wyoming, Geology
 B.S. 2015, Montana State University, Environmental Science, *highest honors*

Appointments:

2023- present	Assistant Professor, University of Connecticut
2022-2023	Postdoctoral Researcher, University of California, Santa Cruz
2021-2022	Postdoctoral Researcher, University of Wyoming
2015-2021	Graduate Research Assistant, University of Wyoming
2017-2018, 2020	Graduate Teaching Assistant, University of Wyoming
2013-2015	Undergraduate Research Assistant, Montana State University

Publications:

Peer-Reviewed Publications [18]:

Tucker, A., Dumont, M., Singley, J.G., Lenssen, N., Callahan, R.P., Marshall, A.M., Jacobsen, L., & Singha, K. (2026). Bridging single-tree processes and landscape-scale patterns to explain vegetation drought resistance and resilience in a Sierra Nevada headwater catchment. *Journal of Geophysical Research: Biogeosciences*. <https://doi.org/10.1029/2025JG009585>

Song, K., Knighton, J., Qiu, S., Yang, X., Suh, J.W., Tavares, J.V., Liu, Y., Tai, X., Fahey, R., Neigh, C.S.R., Callahan, R.P., Hong, F., Li, T., Grinstead, A., Ren, W., Witharana, C., Hedges, S.B., Yang, Z., Vieira Leite, R., Bittencourt, P.R.L., & Zhu, Z. (2026). Physiological fidelity of a satellite-derived forest resilience indicator in the Amazon. *Nature Ecology & Evolution*. <https://doi.org/10.1038/s41559-026-03116-z>

Grana, D., Eppinger, B.J., Flinchum, B.A., Callahan, R.P., Hayes, J.L., Carr, B.J., Moon, S., Parsekian, A.D., Riebe, C.S, Holbrook, W.S. (2026). A rock physics model for geophysical inversion in critical zone studies. *Geophysics*, 91(4), KS129–KS144. <https://doi.org/10.1190/geo-2025-0606>

Donaldson, A., Callahan, R. P., Hahm, W. J., Dralle, D.N., Breunig, R.E., Hudson Rasmussen, B.M., Krone, L.V., Dere, A., Brantley, S.L., West, A.J., Ma, L. Riebe, C.S., Noren, A., & Rempe, D. (2026). Drilling within the critical zone. *Hydrological Processes*, 40(3), e70463. <https://doi.org/10.1002/hyp.70463>

Donaldson, A., Callahan, R. P., Barling, N., Rempe, D., Huang, M.-H., & Zimmer, M. (2026). Neighboring ridgetops with distinct weathering drivers have strikingly similar saprolite chemical weathering. *Journal of Geophysical Research: Earth Surface*, 131(4), e2025JF008348. <https://doi.org/10.1029/2025jf008348>

Bemis, S.P., Holbrook, W.S., Flinchum, B.A., Hayes, J.L., Callahan, R.P., Harman, C.J., Carr, B.J., Riebe, C.S. (2026). Creating a Critical Zone: Feedbacks Between Bedrock Geology, Hydrology, and Vegetation on an exposed Bedrock Surface, Panola Mountain, Georgia, USA. *Journal of Geophysical Research: Earth Surface*, 131(1), e2025JF008424. <https://doi.org/10.1029/2025jf008424>

Wang, L., Warix, S. R., Callahan, R.P., Sullivan, P.L., Singha, K. (2025). Data-model integration to unravel critical zone dynamics: Challenges, successes, and future directions. *WIREs. Water*, 12(6), e70040. <https://doi.org/10.1002/wat2.70040>

Barling, N., Callahan, R. P., Donaldson, A., Huang, M.-H., Rempe, D. M., Shahid, S., Smith, D., Giggy, L., & Zimmer, M. (2025). Greater Rock Moisture Deficits on a Pole-Facing Hillslope Due To Aspect-Related Variations in Vegetation Water Use. *Geophysical Research Letters*, 52(17), e2025GL115750. <https://doi.org/10.1029/2025GL115750>

Flinchum, B.A., Holbrook, W.S., Riebe, C.S, Moon, S., Harman, C.J., Grana, D., Carr, B.J., Hayes, J.L., Neely, A., Callahan, R.P., Richter, D.D., Eppinger, B.J., Uecker, R.K., Chiavello, A., Weinheimer, N.L. (2025). Using seismic refraction data to estimate a relationship between landscape curvature and deep critical zone structure in the South Carolina Piedmont, USA. *Journal of Geophysical Research. Earth Surface*, 130(11). <https://doi.org/10.1029/2025jf008346>

Callahan, R. P., Huang, M.-H., Donaldson, A., Hudson-Rasmussen, B., & Zimmer, M. (2024). Geologic and tectonic controls on deep fracturing, weathering, and water flow in the central California Coast Range. *Geophysical Research Letters*, 51(13). <https://doi.org/10.1029/2024gl109129>

Donaldson, A., Dralle, D., Barling, N., Callahan, R. P., Loik, M. E., & Zimmer, M. (2024). Aspect differences in vegetation type drive higher evapotranspiration on a pole-facing slope in a California oak Savanna. *Journal of Geophysical Research: Biogeosciences*, 129(7), e2024JG008054. <https://doi.org/10.1029/2024jg008054>

Donaldson, A. M., Zimmer, M., Huang, M.H., Johnson, K. N., Hudson-Rasmussen, B., Finnegan, N., Barling, N., & Callahan, R. P. (2023). Symmetry in hillslope steepness and saprolite thickness between hillslopes with opposing aspects. *Journal of Geophysical Research: Earth Surface*, 128(7). <https://doi.org/10.1029/2023jf007076>

Callahan, R. P., Riebe, C. S., Sklar, L. S., Pasquet, S., Ferrier, K. L., Hahm, W. J., Taylor, N. J., Grana, D., Flinchum, B. A., Hayes, J. L., & Holbrook, W. S. (2022). Forest vulnerability to drought controlled by bedrock composition. *Nature Geoscience*, 15(9), 714–719. <https://doi.org/10.1038/s41561-022-01012-2>

Grana, D., Parsekian, A.D., Flinchum, B.A., Callahan, R.P., Smeltz, N.Y., Li, A., Hayes, J.L., Carr, B.J., Singha, K., Riebe, C.S. and Holbrook, W.S. (2022). Geostatistical Rock Physics Inversion for Predicting the Spatial Distribution of Porosity and Saturation in the Critical Zone. *Mathematical Geosciences*. <https://doi.org/10.1007/s11004-022-10006-0>

Riebe, C. S., Callahan, R. P., Granke, S. B.-M., Carr, B. J., Hayes, J. L., Schell, M. S., & Sklar, L. S. (2021). Anisovolumetric weathering in granitic saprolite controlled by climate and erosion rate. *Geology*, 49(5), 551–555. <https://doi.org/10.1130/G48191.1>

Callahan, R. P., Riebe, C. S., Pasquet, S., Ferrier, K. L., Grana, D., Sklar, L. S., Taylor, N. J., Flinchum, B. A., Hayes, J. L., Carr, B. J., Hartsough, P. C., O’Geen, A. T., & Holbrook, W. S. (2020). Subsurface Weathering Revealed in Hillslope-Integrated Porosity Distributions. *Geophysical Research Letters*, 47(15). <https://doi.org/10.1029/2020GL088322>

Leone, J. D., Holbrook, W. S., Riebe, C. S., Chorover, J., Ferré, T. P. A., Carr, B. J., & Callahan, R. P. (2020). Strong slope-aspect control of regolith thickness by bedrock foliation. *Earth Surface Processes and Landforms*, 45(12), 2998–3010. <https://doi.org/10.1002/esp.4947>

Callahan, R. P., Ferrier, K. L., Dixon, J., Dosseto, A., Hahm, W. J., Jessup, B. S., Miller, S. N., Hunsaker, C. T., Johnson, D. W., Sklar, L. S., & Riebe, C. S. (2019). Arrested development: Erosional equilibrium in the southern Sierra Nevada, California, maintained by feedbacks between channel incision and hillslope sediment production. *GSA Bulletin*, 131(7-8), 1179–1202. <https://doi.org/10.1130/B35006.1>

Manuscripts Under Review

Donaldson, A., Callahan, R. P., Belmecheri, S., Nolin, A. F., Hahm, W. J., Dralle, D., Dawson, T. E., Rempe, D. Tree-rings reveal a forest drought vulnerability paradox: larger root-zone storage capacity intensifies drought impacts. In review, *Ecohydrology*.

Weinheimer, N. L., Lukens, C. E., Riebe, C. S., Sklar, L. S., Callahan, R. P., Moon, S., Richter, D. D., Holbrook, W. S., Norton, K. P. Testing for steady state soils in eroding landscapes. In review, *Earth and Planetary Science Letters*.

Manuscripts in Preparation

Callahan, R. P., Riebe, C. S., Sklar, L. S., Moon, S., Carr, B., Grana, D., Singha, K., Flinchum, B., Befus, K., Anderson, S. P. Barnard, H. R., Holbrook, W.S., Microclimatic and topographic stress controls on subsurface weathering in the Colorado Front Range. In prep

Lee, J., Callahan, R.P., Riebe, C., Sklar, L., Weinheimer, N.L., Carr, B., Holbrook, S., Flinchum, B., Uecker, R., Hidy, A., & Moon, S. (2026). Coevolution of deep bedrock weathering and topography in steady-state landscapes. *In preparation*.

Barling, N., Donaldson, A., Callahan, R.P., Rempe, D., Motz, S., & Zimmer, M. (2026). From drought to deluge: Are extreme wet years beneficial to deeply rooted trees? *In preparation*.

Non-refereed Reports and Publications:

Scholz, C. A., Noren, A., Boush, L. P., Carpenter, B. M., & Callahan, R.P. (2026). Drilling down to open up new understanding of Earth's continents. *Eos*. <https://eos.org/articles/drilling-down-to-open-up-new-understanding-of-earths-continents>

Callahan, R. P., Flinchum, B., Aronson, E., Atwood, A., Dere, A., Donaldson, A., Dugan, B., Harman, C., Hayes, J., Hodges, C., Ma, L., Sutfin, N., Michael, H., Person, M., Riebe, C., West, J., Zimmer, M., 2024. Hydrology and Critical Zone Science Planning for Continental Drilling and Coring 2024. <https://sites.google.com/umn.edu/csdscienceplanning/home/hydrology-and-critical-zone-executive-summary>

Jones, C, and many others (including Callahan, R.P.), 2022, Sierra Nevada Field Forum. Field Trip Guidebook. Geological Society of America. Boulder, CO.

Invited Seminars and Lectures:

Geologic and tectonic controls on weathering, water, and ecosystems. Geodynamics Seminar, Lamont-Doherty Earth Observatory, Columbia University, Palisades, NY. March 9, 2026.

Deep critical zone controls on hydrologic processes in mountain landscapes. Distinguished Lecture Series, University of Utah, Salt Lake City, UT. February 19, 2026.

Using near-surface geophysics to understand drivers of hillslope weathering and erosion in mountain landscapes. Guest Lecture Series, University of Massachusetts Amherst, Amherst, MA. November 14, 2025.

The role of the deep critical zone in dictating hydrologic processes in mountain landscapes. Seminar Series, University of Connecticut NRE, Storrs, CT. September 12, 2025.

Lithologic and climatic controls on subsurface weathering and ecosystem dynamics in the Sierra Nevada, California, USA. Tom Oliver Keynote Lecture University of Calgary, Calgary, AB. April 10, 2024.

Digging deeper: quantifying deep weathering to understand ecosystems, water, and erosion in mountain landscapes. University of Connecticut, Storrs, CT. March 2, 2023.

The role of bedrock composition and subsurface weathering in controlling forest drought response in the southern Sierra Nevada, California. University of California, Merced, Merced, CA. January 27, 2023.

The role of deep weathering in ecological and hydrological processes across the mountains of California. University of California, Santa Cruz, Santa Cruz, CA. January 17, 2023.

Bedrock composition and subsurface weathering control forest drought response in the southern Sierra Nevada, California. Clemson University, Clemson, SC. April 29, 2022.

Conference Presentations:

first author presentations [13]:

Callahan, R. P., Donaldson, A., Jarecke, K. M., Price, A., Dumont, M., Lee, J., Quantifying deep subsurface controls on forest dynamics using near-surface geophysics, in-situ measurements, and remote sensing. Abstract B23O-1946, presented at 2025 Fall Meeting, AGU, New Orleans, LA, 16 Dec.

Callahan, R. P., Riebe, C. S., Sklar, L. S., Moon, S., Carr, B., Grana, D., Singha, K., Flinchum, B., Befus, K., Anderson, S. P. Barnard, H. R., Holbrook, W.S., Microclimatic and topographic stress controls on subsurface weathering in the Colorado Front Range. Abstract EP53B-1464, presented at 2024 Fall Meeting, AGU, Washington D.C., 9-13 Dec.

Callahan, R.P., Barling, N., Donaldson, A.M., Huang, MH., Rempe, D.M., Smith, D., Shahid, S., Zimmer, M.A. (2023). Vegetation water-use drives persistent differences in rock moisture content between hillslopes with opposing aspects (**Invited**). Abstract H31E-08, presented at 2023 Fall Meeting, AGU, San Francisco, California, 11-15 Dec.

Callahan, R.P., Huang, MH., Donaldson, A.M., Rempe, D.M., Barling, N.B., Zimmer, M.A. (2023). Deep weathering in the California Coast Range: Implications for water movement and storage in the critical zone, presented at 2023 CUAHSI Biennial Meeting, Tahoe City, California, 11-14 June.

Callahan, R.P., Huang, MH., Donaldson, A.M., Hudson-Rasmussen, B., Rempe, D.M., Schmidt, J., Zimmer, M.A. (2022). Deep critical zone weathering influenced by rock damage from active faults: Implications for catchment water balances. Abstract H22E-08, presented at 2022 Fall Meeting, AGU, Chicago, Illinois, 12-16 Dec.

Callahan, R.P., Grana, D., Holbrook, W.S., Flinchum, B.A., Carr, B.J., Hayes, J.L., Ferrier, K.L., Sklar, L.S., Riebe, C.S. (2021). Interpreting critical zone properties from near-surface geophysics and rock physics modeling: Progress, challenges, and prospects (**Invited**). Abstract EP54B-06, presented at 2021 Fall Meeting, AGU, New Orleans, Louisiana, 13-17 Dec.

Callahan, R.P., Riebe, C.S., Granke, S.B.-M., Carr, B.J., Hayes, J.L., Schell, M.S., & Sklar, L.S. (2020). Anisovolumetric weathering is the norm, not the exception, in granitic saprolite. Abstract EP036-0004, presented at 2020 Fall Meeting, AGU, Virtual, 1-17 Dec.

Callahan, R.P. & Riebe C.S., (2019). Strong lithologic and climatic control on deep subsurface weathering revealed in landscape-scale porosity distributions. Abstract NS23A-05, presented at 2019 Fall Meeting, AGU, San Francisco, California, 9-13 Dec.

Callahan, R.P., Riebe, C.S., Holbrook, W.S., & Goulden, M. (2018). Climatic and lithologic controls on critical zone structure and ecosystem productivity in the Sierra Nevada, California, evaluated using geophysical and geochemical measurements. Abstract EP11D3-2084 presented at 2018 Fall Meeting, AGU, Washington D.C., 10-14 Dec.

Callahan, R.P., Riebe, C.S., & Ferrier, K.L. (2017). Mountain erosion over decades and millennia: new insights from cosmogenic nuclides and sediment yields. Abstract EP33C-0982 presented at 2017 Fall Meeting, AGU, New Orleans, Louisiana, 11-15 Dec.

Callahan, R.P., Taylor, N.J., Pasquet, S., Dueker, K.G., Riebe, C.S., & Holbrook, W.S. (2016). Probing the critical zone using passive-and active-source estimates of subsurface shear-wave velocities. Abstract EP43C-0966, presented at 2016 Fall Meeting, AGU, San Francisco, California, 12-16 Dec.

Callahan, R.P., Riebe, C.S., & Dosseto, A., (2016) Using Cosmogenic and U-Series Nuclides in Stream Sediment to test hypotheses about mountain landscape evolution. Abstract 347, presented at 2016 Goldschmidt Meeting, Yokohama, Japan, Jun 16-Jul 1.

Callahan, R.P. & Hartshorn, A.S. (2014) Soil geochemistry controls fire severity: A soil approach to improved understanding of forest fire consequences in southwest Montana. Abstract GC33E-0571, presented at 2014 American Geophysical Union Fall Meeting, San Francisco, CA 15-19 Dec.

contributing author presentations [36]:

Singha, K., Tucker, A., Dumont, M., Singley, J., Lenssen, N., Callahan, R. P., Marshall, A., Jacobsen, L., Bridging single-tree processes and landscape-scale patterns to explain vegetation drought resistance and resilience in a headwater catchment. Abstract EGU26-262, presented at EGU General Assembly 2026, Vienna, Austria, 6 May.

Moon, S., Lee, J., Callahan, R. P., Riebe, C., Sklar, L., Carr, B., Holbrook, S., Flinchum, B., Weinheimer, N., Uecker, R., Hidy, A., Co-evolution of the critical zone: soil production and bedrock weathering. Abstract EGU26-8745, presented at EGU General Assembly 2026, Vienna, Austria, 6 May.

Singha, K., Tucker, A., Dumont, M., Singley, J. G., Lenssen, N. J. L., Callahan, R. P., Marshall, A. M., Jacobsen, L. L., Multi-scale drivers of intra-catchment variability in drought response: Linking remote-sensing, geophysical, and ecohydrological data. Abstract NS21B-1562, presented at 2025 Fall Meeting, AGU, New Orleans, LA.

Fosdick, J. C., Mota, A., Mooney, J., Mistry, T., Sinoplu, O., Biddle, J., Milanesio, D., Palmero, L., Armas, P., Otamendi, J., Ouimet, W. B., Callahan, R. P., Catchment-wide constraints on long-term erosion, deformation, and weathering to resolve tectonic and climatic controls in the

Sierras Pampeanas (Argentina). Presented at GSA Connects 2025, Geological Society of America, San Antonio, TX, 19 Oct.

Donaldson, A. M., Breunig, R. E., Callahan, R. P., Rempe, D. M., Hahm, W. J., Dralle, D., Dere, A. L. D., Brantley, S., West, A. J., Ma, L., Riebe, C. S., Cantner, K. (2024). An opportunity to identify and discuss best practices for critical zone scientific drilling across disciplines. Abstract H43D-0901, presented at 2024 Fall Meeting, AGU, Washington D.C., 9–13 Dec.

Eligator, L., Hayes, J. L., Uecker, R., Moore, J., Welty, C., Hood, A., Holbrook, S., Flinchum, B., Burgett, D., Eppinger, B., Gupta, S. M., Cummings, W., Berkowitz, A. R., Callahan, R. P., Riebe, C. S. (2024). Critical Zone Science Workshops for K-12 Educators as Catalysts for Curriculum Development and Educator Engagement. Abstract ED11E-2389, presented at 2024 Fall Meeting, AGU, Washington D.C., 9–13 Dec.

Weinheimer, N., Riebe, C.S., Sklar, L.S., Lukens, C.E., Callahan, R.P., Li, T., & Holbrook, S. (2024). Quantifying the response of subsurface weathering to dissection of an uplifting fault block: Insights from cosmogenic nuclides, bulk geochemistry, and seismic refraction surveys. presented at 2024 Fall Meeting, AGU, Washington D.C., 9-13 Dec.

Li, T., Sklar, L.S., Weinheimer, N., Riebe, C.S., Callahan, R.P., Holbrook, S., & Huizenga, C. (2024). Influence of hillslope boulders on the topographic evolution of granitic landscapes. presented at 2024 Fall Meeting, AGU, Washington D.C., 9-13 Dec.

Tucker, A., Dumont, M., Lenssen, N.J.L., Singley, J.G., Callahan, R.P., & Singha, K. (2024). Assessing hydraulic redistribution and tree water use strategies in the Southern Sierra Critical Zone Observatory with geophysical and ecohydrological methods. presented at 2024 Fall Meeting, AGU, Washington D.C., 9-13 Dec.

Shahid, S., Huang, M.-H., Callahan, R.P., Zimmer, M., Barling, N., Giggy, L., & Donaldson, A.M. (2024). Using Bayesian electrical resistivity inversion to reveal hillslope dry-up process in a Mediterranean climate. presented at 2024 Fall Meeting, AGU, Washington D.C., 9-13 Dec.

Joseph, O., Dumont, M., Jarecke, K.M., Hornslein, N., Callahan, R.P., Kusold, K., Blakey, B., Sullivan, P.L., Barnard, H.R., & Singha, K. (2024). Time-lapse electrical resistivity to assess root water uptake of ponderosa pine growing in the Front Range of Colorado. presented at 2024 Fall Meeting, AGU, Washington D.C., 9-13 Dec.

Bemis, S.P., Holbrook, S., Flinchum, B., Aulenbach, B.T., Callahan, R.P., Harman, C.J., Hayes, J.L., Carr, B., & Riebe, C.S. (2024). Hmmm, where to start?: Conditions supporting the initiation of a critical zone on a bedrock surface and feedbacks driving expansion. presented at 2024 Fall Meeting, AGU, Washington D.C., 9-13 Dec.

Lee, J., Moon, S., Callahan, R.P., Riebe, C.S., Hidy, A.J., Carr, B., Flinchum, B., & Uecker, R. (2024). Influence of bedrock weathering on soil production rates in the San Gabriel Mountains, California. presented at 2024 Fall Meeting, AGU, Washington D.C., 9-13 Dec.

Weinheimer, N., Riebe C. S., Callahan, R. P. (2024). Identifying historic land use impacts on soil-mantled landscapes with cosmogenic nuclides: A case study in northwestern South Carolina, presented at AAG Annual Meeting, Honolulu, HI 16-20 Apr.

Lee, J., Moon, S., Riebe, C.S., Callahan, R.P. (2023). Influence of Bedrock Weathering on Soil Production and Hillslope Morphology. Abstract EP51D-164, presented at 2023 Fall Meeting, AGU, San Francisco, California, 11-15 Dec.

Weinheimer, N., Lukens, C.E., Riebe, C.S., Callahan, R.P., Norton, K.P., Moon, S., Sklar, L.S., Neely, A., Richter, D., Schroeder, P.A., Darling, A., Holbrook, W.S. (2023). Soils in the balance: Using cosmogenic nuclides to test for steady-state soil thickness on hillslopes. Abstract EP42C-02, presented at 2023 Fall Meeting, AGU, San Francisco, California, 11-15 Dec.

Flinchum, B., Briglio, J., Riebe, C.S., Callahan, R.P., Sklar, L.S., Holbrook, W.S. (2023). Links between sheet jointing and weathering in saprolite and weathered rock revealed by ground penetrating radar and seismic refraction surveys. Abstract EP23F-1999, presented at 2023 Fall Meeting, AGU, San Francisco, California, 11-15 Dec.

Eligitor, L., Hayes, J.L., Cist, N. Uecker, R., Callahan, R.P., (2023). Connections in the Critical Zone: Bringing Cutting-Edge Science to K-12 Educators. Abstract ED11B-0763, presented at 2023 Fall Meeting, AGU, San Francisco, California, 11-15 Dec.

Donaldson, A.M., Callahan, R.P., Barling, N., Huang, M.H., Rempe, D.M., Zimmer, M.A. (2023). Shallow and deep critical zone controls on water movement and storage. Abstract H31T-1757, presented at 2023 Fall Meeting, AGU, San Francisco, California, 11-15 Dec.

Donaldson, A.M., Barling, N., Callahan, R.P., Motz, S., Rempe, D.M., Zimmer, M.A. (2023). Oak water use patterns between contrasting hydroclimatic and vadose zone storage conditions. Abstract H31E-08, presented at 2023 Fall Meeting, AGU, San Francisco, California, 11-15 Dec.

Weinheimer, N., Riebe, C.S., Callahan, R.P., Neely, A.B. (2022). Steady-state Erosion and Soil Production in the South Carolina Piedmont Inferred from Cosmogenic Nuclide Depth Profiles Abstract EP55C-0839, presented at 2022 Fall Meeting, AGU, Chicago, Illinois, 12-16 Dec.

Kitimikado, C., Carr, B.J., Noren, A.J., Callahan, R.P., Holbrook, W.S., Hayes, J.L., Riebe, C.S. (2022). Understanding the deep critical zone in the South Carolina Piedmont using borehole sampling and imaging. Abstract EP55C-0805, presented at 2022 Fall Meeting, AGU, Chicago, Illinois, 12-16 Dec.

Flinchum, B.A., Holbrook, W.S., Grana, D., Carr, B.J., Callahan, R.P., (2022). Characterizing Deep CZ Structure and Saturation in the Piedmont using P-wave and S-wave Seismic Refraction. Abstract H52H-0542, presented at 2022 Fall Meeting, AGU, Chicago, Illinois, 12-16 Dec.

Holbrook, W.S., Bemis, S.P., Callahan, R.P., Carr, B.J., Flinchum, B.A., Grana, D., Harman, C.J., Hayes, J.L., Moon, S., Neely, A.B., Noren, A.J., Riebe, C.S., Rajaram, H., Richter, D.D., Singha, K., Sklar, L.S. (2021). Controls on critical zone thickness in the Appalachian Piedmont:

lithology, vegetation, and state of stress. Abstract EP43B-06, presented at 2021 Fall Meeting, AGU, New Orleans, Louisiana, 13-17 Dec.

Sklar, L.S., Callahan, R.P., Carr, B.J., Chiavello, A., Cist, N., Davis, E., Flinchum, B.A., Harman, C.J., Hayes, J.L., Holbrook, W.S., Litwin, D.G., Moon, S., Neely, A.B., Plante, Z., Richter, D.D., Riebe, C.S., Singha, K., Weinheimer, N. (2021). Variation in Hillslope Sediment Size Controlled by Differences in Subsurface Weathering in a Transient Piedmont Landscape, South Carolina, USA. Abstract EP45G-1573, presented at 2021 Fall Meeting, AGU, New Orleans, Louisiana, 13-17 Dec.

Bemis, S.P., Holbrook, W.S., Hayes, J.L., Harman, C.J., Flinchum, B.A., Callahan, R.P., Riebe, C.S., Carr, B.J. (2021). Growing an Oasis: The Role of Trees in Driving Regolith Production in a Bedrock Landscape, Panola Mountain, Georgia. Abstract EP45G-1573, presented at 2021 Fall Meeting, AGU, New Orleans, Louisiana, 13-17 Dec.

Flinchum, B.A., Holbrook, W.S., Eppinger, B.J., Hayes, J.L., Richter, D.D., Harman, C.J., Callahan, R.P., Moon, S., Singha, K., Ferguson, T.A., Carr, B.J., Bemis, S.P. (2021). Connecting the deep and shallow critical zone: Quantifying inherited heterogeneity of the CZ structure. Abstract EP45G-1585, presented at 2021 Fall Meeting, AGU, New Orleans, Louisiana, 13-17 Dec.

Brigilio, J., Flinchum, B.A., Callahan, R.P., Riebe, C.S., Holbrook, W.S. (2021), Differing Subsurface Fracture Characteristics Inferred from Ground Penetrating Radar at Two Granitic Sites in the Southern Sierra Nevada, California. Abstract EP55G-1182, presented at 2021 Fall Meeting, AGU, New Orleans, Louisiana, 13-17 Dec.

Harman, C.J., Bemis, S.P., Callahan, R.P., Carr, B.J., Eppinger, B.J., Flinchum, B.A., Hayes, J.L., Holbrook, W.S., Litwin, D.G., Moon, S., Riebe, C.S., Singha, K., Sklar, L.S. (2021). Panola Mountain revisited: intensive geophysical and geochemical studies reveal the structure of the deep critical zone at a classic hydrologic study site Abstract H41B-06, presented at 2021 Fall Meeting, AGU, New Orleans, Louisiana, 13-17 Dec

Hayes, J.L., Carr, B.J., Holbrook, W.S., Callahan, R.P., Harman, C.J., Riebe, C.S., Bemis, S.P., Flinchum, B.A., Moon, S., Singha, K., Sklar, L.S. (2021). Geophysical imaging reveals lithologic controls on weathering at the Panola Mountain Research Watershed, Georgia, USA Abstract NS35D-0396, presented at 2021 Fall Meeting, AGU, New Orleans, Louisiana, 13-17 Dec

Del Vecchio, J., Hassenruck-Gudipati, H.J., Roth, D.L., Merritts, D., Hill, K.M., Sun, X., Kwang, J., Koppes, M.N., Mahon, R., Madoff, R., Gaspirini, N.M., Lehnigk, K., McDowell, C., Callahan, R.P., Mukherjee, U., Sklar, L.S., Gagliardi, J., Luna, L., Straub, K.M. (2021). URGE Pod Outcomes for the AGU EPSP Section. Abstract U35A-2265, presented at 2021 Fall Meeting, AGU, New Orleans, Louisiana, 13-17 Dec

Gaspirini, N.M., Roth, D.L., Madoff, R., Mukherjee, U., Callahan, R.P., Mahon, R., Sklar, L.S., Gagliardi, J., Lehnigk, K., Luna, L., Merritts, D., Kwang, J., Del Vecchio, J., Sun, X., Koppes,

M.N., McDowell, C., Straub., K.M., Hassenruck-Gudipati, H.J. (2021). Lessons learned from the AGU EPSP URGE pod on how to structure an equitable, inclusive, and safe committee space. Abstract U35A-228, presented at 2021 Fall Meeting, AGU, New Orleans, Louisiana, 13-17 Dec

Brigilio, J., Flinchum, B.A., Callahan, R.P., & Riebe, C.S. (2020). Ground Penetrating Radar Reveals Differences in Fracture Spacing Across Two Geochemically Distinct Granitic Lithologies. Abstract NS014-0009 presented at 2020 Fall Meeting, AGU, Virtual, 1-17 Dec.

Riebe, C.S. & Callahan, R.P., (2019). Mountain ecosystem response to drought moderated by lithologic and climatic controls on subsurface water-storage capacity. Abstract B22E-03, presented at 2019 Fall Meeting, AGU, San Francisco, California, 9-13 Dec.

Riebe, C.S. & Callahan, R.P., (2019). Lithologically mediated feedbacks between subsurface weathering and ecosystem productivity. Abstract 2831, presented at 2019 Goldschmidt Meeting, Barcelona, Spain 18-23 August. (R.P. Callahan served as speaker)

Riebe, C.S., Callahan, R.P., & Arvin L.J, (2018). Built on bedrock, running on dust: Controls on ecosystem productivity and vulnerability in the Sierra Nevada, California. Abstract 2157, presented at 2018 Goldschmidt Meeting, Boston, Massachusetts, 12-17 August.

Riebe, C.S., Callahan, R.P., Goulden M.L., Pasquet S., Flinchum B.A., Taylor N.J., & Holbrook W.S. (2017). The influence of subsurface porosity and bedrock composition and ecosystem productivity in the Sierra Nevada Batholith. Abstract EP53D-1762 presented at 2017 Fall Meeting, AGU, New Orleans, Louisiana, 11-15 Dec.

Taylor, N. J., Riebe, C. S., Dueker, K. G., Goulden, M., Flinchum, B. A., Pasquet, S., Callahan, R.P., Hahm, W.J. & Holbrook, W. S. (2016). Comprehensive seismic surveys suggest that subsurface water-holding capacity is secondary to bedrock nutrient content as a regulator of vegetation productivity in the Sierra Nevada Batholith, California. Abstract EP43C-0964, presented at 2016 Fall Meeting, AGU, San Francisco, California, 12-16 Dec.

Teaching Experience:

Lead Instructor at UConn:

Course	Semester/Year	Credits	Enrollment
Creating and Sustaining National Parks (ERTH-2310E)	Fall 2023	3	45
Independent Study (ERTH-4999)	Fall 2023	1	1

Engineering & Environmental Geology (ERTH 3710/ENVE 3530)	Spring 2024	3	22
Independent Study (ERTH-4999)	Spring 2024	1	1
Creating and Sustaining National Parks (ERTH-2310E)	Fall 2024	3	91
Independent Study (ERTH-4999)	Fall 2024	1	2
Engineering & Environmental Geology (ERTH 3710/ENVE 3530)	Spring 2025	3	18
Independent Study (ERTH-4999)	Spring 2025	1	1
Applied and Environmental Geophysics (ERTH 4510/5510)	Fall 2025	3	8
Independent Study (ERTH 4999)	Fall 2025	1	2
Engineering & Environmental Geology (ERTH 3710/ENVE 3530)	Spring 2026	3	24

Graduate Student Advising:

Karimah Comstock (M.S. student, University of Connecticut) 2024-present
Project: Subsurface weathering and ecosystem dynamics in the southern Sierra Nevada, California

Joshua Lee (Ph.D. student, University of Connecticut) 2024-present
Project: Geologic controls on water storage capacity and ecosystem dynamics in the volcanic landscape of eastern Oregon

Graduate Student Committee Membership: Kevin Li (University of Connecticut), Caroline Stanton (University of Connecticut), Arielle Mota (University of Connecticut), David Luffman (University of Connecticut), Victor Oladoja (University of Connecticut), Piper Lovegreen (University of California, Santa Barbara)

Previous Graduate Student Mentoring:

Nerissa Barling (Ph.D. student, University of California, Santa Cruz) 2022-2026
Project: Variations in moisture content in hillslopes with opposing aspects

Amanda Donaldson (Ph.D. student, University of California, Santa Cruz) 2022-2024
Project: Microclimate controls on subsurface weathering in the central California Coast Range

Sarah Granke (MS student, University of Wyoming) 2019-2020
Project: Quantifying physical and chemical weathering saprolite of the southern Sierra Nevada, California.

Undergraduate Student Mentoring:

Sean Kick (University of Connecticut) 2025
Project: Subsurface critical zone structure in western US

Adam Agiro (University of Connecticut) 2024-2025
Project: Subsurface structure of Horsebarn Hill

Michael Magut (University of Connecticut) 2023-2024
Project: Stream intermittency in UConn Forest

Allison Bijas (University of Connecticut) 2023
Project: Topographic controls on subsurface weathering in the southern Sierra Nevada, California

Bryanna Pilkington (Concordia University, Montreal, Canada) 2020-2021
Project: Climatic, lithologic, and weathering controls on forest drought response in the San Jacinto and San Gabriel Mountains, California.

Andrew Miller (University of Wyoming) 2018-2019
Project: Geochemical characterization of subsurface weathering in the southern Sierra Nevada, California.

Grants & Contracts:

Funded:

Title of Project: Investigation of hillslope hydrologic connectivity of valley-bottom and upland-restoration

Agency/Company: US Forest Service

Total Dollar Amount: \$55,000

Role: PI

Period of Contract: 07/2024-10/2027

Title of Project: Geophysical Mapping of the Subsurface Critical Zone for Improved Land Surface Hazard Assessment

Agency/Company: NSF Center for Land Surface Hazards (CLaSH)

Total Dollar Amount: \$35,000

Role: PI

Period of Contract: 05/2026-04/2027

Pending:

Title of Project: The Role of Critical Zone Water Storage Capacity in Regulating Vegetation Response to Drought Across Timescales

Agency/Company: National Science Foundation

Total Dollar Amount: \$614,123

Role: PI

Period of Contract: 01/2027-12/2029

Title of Project: Deep Subsurface Controls on Forest Productivity and Regeneration

Agency/Company: US Department of Agriculture

Total Dollar Amount: \$284,290

Role: PI

Period of Contract: 06/2026-06/2028

Not Funded:

Title of Project: Controls of bedrock weathering and water storage capacity on vegetation water use strategies, structural overshoot, and drought resilience in California

Agency/Company: National Science Foundation

Total Dollar Amount: \$539,160

Role: PI

Collaborators: Michael Hren (Co-PI)

Period of Contract: 05/2025-05/2028

Title of Project: Collaborative Research: CAIG: Headwaters AI: Transforming hydrologic prediction in headwaters using AI and synthesized subsurface structure

Agency/Company: National Science Foundation

Total Dollar Amount: \$726,648

Role: Co-I

Period of Contract: 01/2027-12/2029

Honors & Awards:

2026 Fiona Kirkby Award Recipient (For excellence in reviewing for *Earth Surface Processes and Landforms*)

Professional Service:Professional Service:

-Member Continental Scientific Drilling Office Executive Board

-Participant in AGU Earth and Planetary Surface Processes URGE pod

Workshop Organization:

-Co-chair of Continental Scientific Drilling Community Science Planning Hydrology and Critical Zone Science working group.

Conference and Field Trip Organization:

-Convenor: AGU 2021 EP041 Top-down and Bottom-up Controls on Critical Zone Structure, Processes, and Resilience; AGU 2024 EP007: Bedrock breakdown: The role of weathering in the evolution and function of the critical zone; AGU 2024 H118 - Rock Moisture in Weathered Bedrock: Understanding the Hydrologic, Ecological, and Geochemical Function of Water Movement and Storage Within Weathered Bedrock; AGU 2025 EP41B-From Bedrock to Treetop: Hillslope Processes, Weathering, and the Evolution of the Critical Zone Across Scales

-Participant and stop leader in 2022 GSA Thompson Field: “Old or Young the Topographic Evolution of the Sierra Nevada.

Peer Review:

-Journal Reviewer: *Earth Surface Processes and Landforms*, *Proceedings of the National Academies of Sciences*, *Agricultural and Forest Meteorology*, *Nature Geoscience*, *American Journal of Science*; *Geology*, *GSA Field Guides*, *Water Resources Research*

Proposal Review:

-Ad-hoc Proposal Reviewer: *National Science Foundation Hydrological Sciences*, *National Science Foundation Water, Landscape, and Critical Zone Processes*, *German Research Foundation*

Community Outreach:

-Co-leader Critical Zone High School Teacher Workshop (July 2025). I co-led a workshop that focused on implementing critical zone science into high school curricula. The workshop had 11 participants from across Connecticut participate.

University/Department Service:

-Member UConn Environmental Science Executive Board
-Member University of Connecticut Department of Earth Sciences DEI committee

Membership in Professional Societies:

American Geophysical Union
Geological Society of America